

- 2(a) By the principle of conservation of momentum, since there is no external force acting, the total change in momentum of the of ball and wall = 0. [1]

Therefore the change in momentum (impulse) of the wall is equal and opposite to the change in momentum of the ball . [1]

Therefore, Student A is wrong.

- (b) Taking values from Fig 2.2,
Total momentum before collision = $(1.2 \times 4) + 0 = 4.8 \text{ kg m s}^{-1}$ [1]
Total momentum after collision = $(1.6 \times 3.6) + (-0.8 \times 1.2) = 4.8 \text{ kg m s}^{-1}$ [1]

Since total momentum before collision is equal to the total momentum after collision, momentum is conserved in this collision. [1]

- (c) Relative speed of approach = $4.0 - 0 = 4.0 \text{ m s}^{-1}$
Relative speed of separation = $1.6 - (-0.8) = 2.4 \text{ m s}^{-1}$ [1]

Since relative speed of approach is not equal to relative speed of separation, the collision is inelastic. [1]